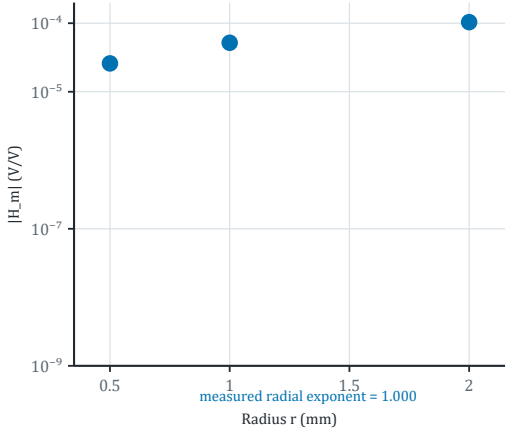
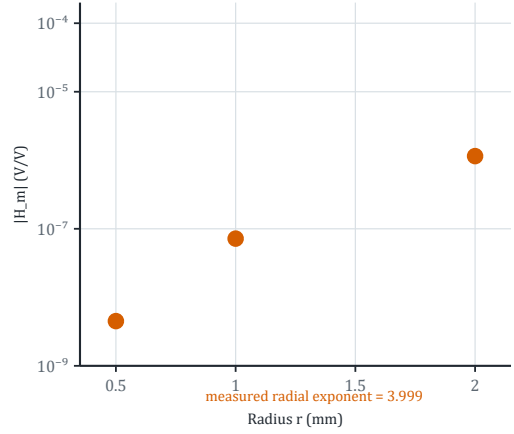
Symmetry-breaking dipole and nominal octupole harmonics

Complex-potential Fourier magnitudes on deterministic rings; radial-out displacement, h50

A m = 1 · induced dipole



B m = 4 · nominal octupole



In the local harmonic expansion, $\Phi_1 \propto r$ gives an approximately constant dipolar field; $\Phi_4 \propto r^4$ gives $E_4 \propto r^3$.

At r = 0.5, 1 and 2 mm, the m = 1 to m = 4 electric-field amplitude ratio is about 1.44×10^3, 1.80×10^2 and 22.5.